

A NONDEGENERATE ELLIPTIC ISLAND FOR THE $(1, 1)$ SYMMETRIC KICKED HARPER

ELMAR HAMILIJAONA

ABSTRACT. We give a computer-assisted proof that the symmetric kicked Harper map at parameters $(1, 1)$ has a twist-nondegenerate elliptic periodic orbit of least period six and rotation vector $(1/2, 0)$. Both the orbit and the non-zero first Birkhoff exponent are certified using the methodology of interval arithmetic. Moser's local invariant curve theorem then provides positive-measure sets consisting of invariant KAM-type curves for the sixth iterate. We prove directly that Lyapunov characteristic exponents vanish at every point of these curves (and of their invariant saturation) under our original map. In particular, the map is not ergodic with respect to area and is not nonuniformly hyperbolic almost everywhere.

1. INTRODUCTORY REMARKS

Historically, dynamical systems developed in close connection with mechanics and physics. In the late nineteenth and early twentieth centuries, the work of Ludwig Boltzmann, James Clerk Maxwell, and Josiah Willard Gibbs helped shape the statistical-mechanical questions from which ergodic theory developed. A central problem was to justify identifying time averages with phase-space averages. Boltzmann's ergodic hypothesis and the later quasi-ergodic hypothesis were important stages in this history, but orbit density alone is not the modern measure-theoretic condition of ergodicity; see Moore's historical account [Moo15].

The foundational ergodic theorems are Birkhoff's pointwise theorem (1931) and von Neumann's mean theorem (1932) [Moo15]. In their modern forms, the former gives almost-everywhere convergence for integrable observables, while the latter gives convergence in L^2 for square-integrable observables. The limits are invariant functions; ergodicity is the additional hypothesis that makes them equal, almost everywhere, to the corresponding phase-space averages. Thus the ergodic theorems do not by themselves establish ergodicity of a particular mechanical system.

Hamiltonian mechanics provides another central source of questions in deterministic dynamics. Kolmogorov–Arnold–Moser theory describes the persistence of many quasiperiodic invariant tori under sufficiently small perturbations of integrable systems, subject to arithmetic and nondegeneracy hypotheses (see [Sin88], [Mos62], [Mos].) Hyperbolic dynamics approaches stability and instability through invariant contracting and expanding directions. As Pesin emphasizes in his contribution to the volume edited by Sinai, this theory connects local properties of trajectories to global dynamical behavior [Sin88, pp. 108–151].

These viewpoints are complementary: a positive-area region of regular motion need not exclude hyperbolic behavior elsewhere.

The kicked Harper model provides a setting for these questions. It is closely related to the kicked-rotor paradigm and the Chirikov standard map, and its quantum version attracted interest because dynamical localization can fail [Art11]. Here we study the classical area-preserving map, not quantum localization or spectral properties.

Kicked Harper model. In angular canonical variables (q, p) , a conventional kicked Hamiltonian is

$$\mathcal{H}(q, p, t) = \ell \cos p + \kappa \cos q \sum_{j \in \mathbb{Z}} \delta(t - j); \quad (1)$$

see [Art11]. Here $\delta(t - j)$ is the Dirac distribution at j . It represents an instantaneous kick of unit integrated weight. The first term is the momentum-dependent part of the Hamiltonian and the second is the periodically applied kicking potential.

With the canonical convention

$$\dot{q} = \frac{\partial \mathcal{H}}{\partial p}, \quad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q},$$

we have that (just before consecutive kicks):

$$p_{n+1} = p_n + \kappa \sin q_n, \quad q_{n+1} = q_n - \ell \sin p_{n+1}.$$

Indeed, the kick first changes p while keeping q unchanged, then the following unit-time evolution changes q while keeping the momentum p fixed. To state the precise torus convention used below, set

$$x = \frac{q}{2\pi}, \quad y = \frac{p}{2\pi}, \quad K = -\frac{\ell}{2\pi}, \quad L = \frac{\kappa}{2\pi}.$$

This explicit rescaling and parameter relabeling give the convention of Jäger, Koropecski, and Tal [JKT21]:

$$F_{K,L}(x, y) = (x + K \sin(2\pi(y + L \sin(2\pi x))), y + L \sin(2\pi x)). \quad (2)$$

Throughout the remainder of the paper, the parameters are (K, L) as in (2).

Let $\mathbb{T}^2 = \mathbb{R}^2 / \mathbb{Z}^2$, with normalized Lebesgue measure m . The map is the composition $F_{K,L} = H_K \circ V_L$, where

$$V_L(x, y) = (x, y + L \sin(2\pi x)), \quad H_K(x, y) = (x + K \sin(2\pi y), y).$$

Both shears are real-analytic diffeomorphisms with determinant one. Moreover,

$$F_{K,L}(z + k) = F_{K,L}(z) + k \quad (z \in \mathbb{R}^2, k \in \mathbb{Z}^2). \quad (3)$$

Consequently, $F_{K,L}$ induces a real-analytic area-preserving torus diffeomorphism $\Phi_{K,L}$, isotopic to the identity.

The symmetric case in this convention is $K = L$. Direct calculations show the following important relation related to sign convetions: for $T(x, y) = (x, y + 1/2)$, direct substitution gives $T^{-1} \circ F_{K,L} \circ T = F_{-K,L}$.

Artuso describes a progression from an integrable regime toward strongly chaotic behavior as the symmetric parameter is increased, with mixed regime at intermediate values [Art11]. This is physical motivation, not a theorem that all sufficiently large parameters yield Lebesgue ergodicity or have no elliptic islands. The same source explains the Floquet viewpoint: because the quantum Hamiltonian is periodic in time, its one-period unitary propagator governs the stroboscopic quantum evolution. These connections to physics motivate the family without entering the classical proof below.

1.1. Hyperbolicity of smooth dynamical systems in the sense of Pesin. We restrict to discrete-time dynamical systems, or cascades. The background terminology follows Barreira–Pesin, Katok–Hasselblatt, and the hyperbolicity chapter in the volume edited by Sinai [BP02, KH95, Sin88]. It is important to distinguish a uniformly hyperbolic set, a nonuniformly hyperbolic measure, and the weaker existence of a positive-measure hyperbolic component.

Definition 1.1 (Hyperbolic Sets; see [BS02]). *Let M be a C^1 compact Riemannian manifold, $\varphi : M \rightarrow M$ a C^1 diffeomorphism. A compact φ -invariant subset $\Lambda \subset M$ is called (uniformly)hyperbolic if there exists (a) constants $C > 0$, $\lambda \in (0, 1)$; (b) families of $D\varphi$ -invariant subspaces $E^s(x) \subset E^u(x) \subset T_x M$, $x \in \Lambda$ such that, for every $x \in \Lambda$:*

- (i) $T_x M = E^s(x) \oplus E^u(x)$;
- (ii) $\|D\varphi^n(x)v^s\| \leq C\lambda^n\|v^s\|$ for every $v^s \in E^s(x)$ and $n \geq 0$;
- (iii) $\|D\varphi^{-n}(x)v^u\| \leq C\lambda^n\|v^u\|$ for every $v^u \in E^u(x)$ and $n \geq 0$.

If $\Lambda = M$, then φ is an Anosov diffeomorphism. The second inequality expresses backward contraction in the unstable direction, equivalently forward expansion. Thus the stable and unstable directions have opposite time orientations. Uniform partial hyperbolicity allows a center bundle between uniformly contracting and expanding bundles, together with domination of their rates: $T_x M = E^s(x) \oplus E^c(x) \oplus E^u(x)$; in particular, it is a different notion from Definition 1.1.

For an invariant Borel probability measure μ , Oseledets’ theorem gives a full-measure set of regular points with a Lyapunov splitting. Its integrability assumptions hold here because M is compact and φ and φ^{-1} are C^1 . At such a point, for every $v \neq 0$, the directional exponent is

$$\chi_\varphi(z, v) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|D\varphi^n(z)v\|.$$

An individual nonregular point need not have this limit. The later argument on KAM curves establishes the limit directly at every point, rather than merely almost everywhere.

The definitions of nonuniform complete and partial hyperbolicity appear in [BP02, Sin88]. We use the following equivalent (in our specific settings) formulation:

Definition 1.2 (Nonuniform complete hyperbolicity and the measure convention). *A dynamical system is referred to as nonuniformly completely hyperbolic (NCH) if all its Lyapunov exponents are non-zero.*

An invariant probability measure μ is hyperbolic if this holds μ -almost everywhere. We say that a map is nonuniformly hyperbolic almost everywhere with respect to m when m is a hyperbolic measure.

In formulations that call a system nonuniformly completely hyperbolic when it has an invariant set of *positive* m -measure consisting of such trajectories, the conclusion is weaker. A positive-measure set of zero exponents rules out the almost-everywhere property, but does not rule out this positive-measure version. Our conclusion concerns the almost-everywhere property.

Proposition 1.3. *At a Lyapunov-regular point of an area-preserving surface diffeomorphism, the two exponents satisfy*

$$\lambda_+(z) + \lambda_-(z) = 0, \quad \lambda_+(z) \geq \lambda_-(z).$$

The point is completely hyperbolic exactly when $\lambda_+(z) > 0 > \lambda_-(z)$, equivalently when $\chi_\varphi(z, v) \neq 0$ for every $v \neq 0$.

Proof. In area coordinates, $\det D\varphi^n(z) = 1$, so the sum of the exponents is zero. The equivalence follows from the Lyapunov splitting at a regular point. $\square$

Nonuniform partial hyperbolicity allows a center part alongside contracting or expanding directions; the precise splitting and rate hypotheses are essential [BP02, Sin88]. Finiteness of an exponent alone does not express this property: on a compact manifold all existing directional exponents of a C^1 diffeomorphism are finite, since $D\varphi$ and $D\varphi^{-1}$ are bounded. In [Sin88, Theorem 2.12, pp.123], a system is nonuniformly partially hyperbolic (NPH) if, for every trajectory, there exists a point where the characteristic Lyapunov exponent is negative.

1.2. KAM theory and positive-area invariant sets. The KAM viewpoint suggests a mechanism for non-ergodicity: suitable perturbations of nondegenerate integrable Hamiltonian systems retain positive-measure families of invariant tori [Sin88, Mos]. In the two-dimensional setting, the relevant objects are invariant circles. Smallness, twist, and arithmetic hypotheses are indispensable; the conclusion does not apply to an arbitrary perturbation of an arbitrary integrable system.

On the annulus $\mathbb{T}^1 \times \mathbb{R}$, an area-preserving diffeomorphism ψ is exact symplectic when $\psi^*(y d\theta) - y d\theta$ is exact. Equivalently, the algebraic area between an essential simple loop and its image is zero [JKT21, Section 2.3].

Proposition 1.4 (Invariant-curve theorem; [JKT21, Theorem 2.8]). *Let $\psi_0(\theta, I) = (\theta + \omega(I), I)$, where $\omega : \mathbb{R} \rightarrow \mathbb{R}$ is smooth. Suppose $\omega'(I_0) \neq 0$ and $\alpha = \omega(I_0) \bmod 1$ is Diophantine: for some $\gamma > 0$ and $\nu > 1$,*

$$|q\alpha - p| \geq \gamma|q|^{-\nu} \quad (p \in \mathbb{Z}, q \in \mathbb{Z} \setminus \{0\}), \quad (4)$$

using any real representative of α . Every smooth exact symplectic diffeomorphism ψ sufficiently close to ψ_0 in the C^∞ topology has a smooth invariant circle, a graph $I = u_\psi(\theta)$ near I_0 , on which ψ is topologically conjugate to the rotation by α . The invariant graphs can be chosen to depend continuously on ψ .

This single-circle statement explains the persistence mechanism but, by itself, does not assert that the union of persistent curves has positive area or that their conjugacies are differentiable. For those conclusions we use the stronger local KAM formulation near a twist-nondegenerate elliptic point in Section 5. The computation in this paper establishes its hypotheses at $(K, L) = (1, 1)$; it does not produce explicit parametrizations of the invariant curves.

Rotation vector of a lifted orbit. For a torus diffeomorphism isotopic to the identity, let $\mathcal{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a lift satisfying integer equivariance. The rotation vector at z is

$$\rho(\mathcal{F}, z) = \lim_{n \rightarrow \infty} \frac{\mathcal{F}^n(z) - z}{n},$$

when the limit exists. The chosen lift is part of the definition [JKT21].

Proposition 1.5. *If $\mathcal{F}^q(z) = z + v$ with $q \geq 1$ and $v \in \mathbb{Z}^2$, then $\rho(\mathcal{F}, z) = v/q$.*

Proof. Integer equivariance gives $\mathcal{F}^{kq}(z) = z + kv$ by induction. For $n = kq + r$, $0 \leq r < q$, it follows that

$$\mathcal{F}^n(z) - z = kv + \mathcal{F}^r(z) - z.$$

The last term takes only finitely many values. Divide by n and use $k/n \rightarrow 1/q$. $\square$

Resonance for vector fields and maps. The normal-form viewpoint of the volume edited by Anosov and Arnold in the EMS volume (see Dynamical System I, [AA88]) provides useful background. For a vector field with linear eigenvalues $\alpha = (\alpha_1, \dots, \alpha_d)$, a nonlinear monomial $z^k e_j$ is resonant when

$$\alpha_j = \langle k, \alpha \rangle = \sum_{i=1}^d k_i \alpha_i, \quad k_i \in \mathbb{Z}_{\geq 0}, \quad |k| \geq 2. \quad (5)$$

The linear vector field is called resonant if such a relation occurs, and nonresonant otherwise. The sum runs over all indices, including $i = j$.

For a discrete map, the corresponding relation is multiplicative. If its linear multipliers are $\lambda_1, \dots, \lambda_d$, the monomial $z^k e_j$ is resonant when (see [AA88, pp.99])

$$\lambda_j = \prod_{i=1}^d \lambda_i^{k_i}. \quad (6)$$

Indeed, the homological equation for a polynomial change of coordinates has divisor $\prod_i \lambda_i^{k_i} - \lambda_j$, whereas the vector-field equation has divisor $\langle k, \alpha \rangle - \alpha_j$. This distinction explains how the classical resonance discussion in [AA88] enters the present discrete-time problem.

In the elliptic surface case, the pair is $(\lambda, \bar{\lambda})$ with $|\lambda| = 1$. A monomial $z^j \bar{z}^k$ in the first component is resonant when

$$\lambda^j \bar{\lambda}^k = \lambda, \quad \text{or equivalently} \quad \lambda^{j-k-1} = 1.$$

The monomial $z^2 \bar{z}$ is always resonant. The conditions $\lambda^r \neq 1$ for $1 \leq r \leq 4$ remove the other quadratic and cubic obstructions; they do not assert nonresonance at every order.

The coefficient remaining at $z^2\bar{z}$, after quadratic elimination and symplectic normalization, is the first Birkhoff or twist coefficient used below.

2. MAIN RESULTS

Throughout the proof, set

$$F = F_{1,1}, \quad f = \Phi_{1,1}, \quad P = F^6 - (3, 0), \quad g = f^6.$$

Thus P is a lift of g . We keep P and g distinct: P acts on the plane, whereas g acts on the torus.

The rotation-theoretic work of [JKT21] describes the onset of diffusion in this family. In particular, the rotation set has nonempty interior on the nonzero diagonal of parameter space. That conclusion does not assert Lebesgue ergodicity and does not exclude local elliptic islands. The contribution here is an explicit, rigorously validated elliptic six-cycle and its nonzero twist at the particular parameter $(1, 1)$.

Theorem 2.1. *There exists $p \in \mathbb{R}^2$ with the following properties.*

- (1) $F^6(p) = p + (3, 0)$, and the projection $\bar{p} \in \mathbb{T}^2$ has least period six under f . Its rotation vector, relative to the lift F , is $(1/2, 0)$.
- (2) The fixed point p of P is elliptic. Its multipliers satisfy $\lambda^j \neq 1$ for $1 \leq j \leq 4$, and its first Birkhoff coefficient is nonzero. In the symplectic convention specified in Section 5,

$$-168234731311.063214153 < \tau_{\text{can}} < -168234731311.063214152. \quad (7)$$

- (3) Every sufficiently small neighborhood U of (the projected point) $\bar{p}$ contains a measurable set K of positive area, consisting of smooth g -invariant Jordan curves on which g is smoothly conjugate to Diophantine circle rotations. At every point z of

$$\widehat{K} = \bigcup_{j=0}^5 f^j(K)$$

and for every $0 \neq v \in T_z \mathbb{T}^2$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \|Df^n(z)v\| = 0. \quad (8)$$

- (4) K can be chosen so that $0 < m(\widehat{K}) < 1$. Hence f is not ergodic with respect to m , and its Lyapunov exponents are not both nonzero m -almost everywhere.

Here and below, a nonzero first Birkhoff coefficient means *twist nondegeneracy*. The numerical coefficient depends on the stated coordinate convention; its nonvanishing does not. No assertion that the central multiplier is irrational or Diophantine is needed.

The proof separates finite computation from local dynamics. Section 3 validates the periodic orbit and its multipliers. Section 4 validates the twist. Sections 5 and 6 give the KAM and measure-theoretic consequences. Appendix A describes the arithmetic and the complete reproducible checker.

3. VALIDATION OF THE PERIODIC ORBIT

Define

$$G(z) = F^6(z) - z - (3, 0) = P(z) - z. \quad (9)$$

All finite decimal constants in the proof and the checker are interpreted as exact rational numbers, not binary floating-point approximations.

3.1. Exact input data and the Krawczyk test. Use the center $c = (c_x, c_y)$ with the following full decimal expansions:

$$c_x = 0.27474448332000881878050615161797640957394247787842890490043558664838204815059897, \\ c_y = 0.092588780576427956472961292120881745888037901834466710224563523023949661988119064.$$

For $r > 0$, let

$$X_r = c + [-r, r]^2.$$

The fixed rational preconditioner C_0 has entries

$$(C_0)_{11} = 0.4135021217575490703333801035742412075797, \\ (C_0)_{12} = -2.399328010952348941913855704225548696173, \\ (C_0)_{21} = 0.3480268316536689025396153232872661506545, \\ (C_0)_{22} = -1.413502121757549070333380103574241207580.$$

The checker verifies $\det C_0 \neq 0$ using exact rational arithmetic. No interval matrix inversion is required.

Let $[DG](X_r)$ denote an interval matrix containing $DG(z)$ for every $z \in X_r$. Set

$$\mathcal{K}(c, X_r) = c - C_0 G(c) + (I - C_0 [DG](X_r))(X_r - c). \quad (10)$$

The following elementary contraction version of the Krawczyk criterion suffices; see also [Kra69].

Lemma 3.1. *Suppose C_0 is nonsingular,*

$$\mathcal{K}(c, X_r) \subset \text{int } X_r, \quad \|I - C_0 [DG](X_r)\|_\infty < 1.$$

Then G has exactly one zero in X_r .

Proof. Set $T_0(z) = z - C_0 G(z)$. The integral mean-value formula gives

$$T_0(z) \in c - C_0 G(c) + (I - C_0 [DG](X_r))(z - c) \subset \mathcal{K}(c, X_r) \quad (z \in X_r).$$

Thus T_0 maps the closed box X_r into itself. The derivative bound makes T_0 a contraction in the maximum norm. The contraction mapping theorem gives a unique fixed point, and the nonsingularity of C_0 makes fixed points of T_0 exactly zeros of G . $\square$

For completeness, the one-step derivative used in the interval propagation is

$$DF(x, y) = \begin{pmatrix} 1 + ab & a \\ b & 1 \end{pmatrix}, \quad \begin{cases} b = 2\pi \cos(2\pi x), \\ a = 2\pi \cos(2\pi(y + \sin(2\pi x))). \end{cases} \quad (11)$$

Its determinant is exactly one. The checker propagates both the state and the derivative through six iterates, then uses $DG = DF^6 - I$.

Proposition 3.2 (Validated root). *The map G has a unique zero in $X_{10^{-30}}$. This zero belongs to $X_{10^{-60}}$.*

Proof. The outward-rounded interval calculation gives the strict upper bounds in Table 1. Each row satisfies Lemma 3.1. The boxes have the same exact center and are nested, so their zeros are the same point. The listed bounds are checked by exact rational comparisons

TABLE 1. Certified strict upper bounds, valid in the recorded runs at 256, 384, and 512 binary fractional bits. The final column bounds the maximum absolute coordinate of $\mathcal{K}(c, X_r) - c$.

r	$\ I - C_0[DG](X_r)\ _\infty$	$\ \mathcal{K}(c, X_r) - c\ _\infty$
10^{-30}	10^{-18}	10^{-48}
10^{-48}	10^{-36}	10^{-64}
10^{-60}	10^{-38}	10^{-64}

in the executable verifier, not inferred from rounded output. $\square$

3.2. Least period and rotation vector. Since $G(p) = 0$, we have $F^6(p) = p + (3, 0)$. Let d be the least period of $\bar{p}$ under f . Then $d \mid 6$, and $F^d(p) = p + v$ for some $v \in \mathbb{Z}^2$. By (3), induction gives

$$F^{kd}(p) = p + kv \quad (k \geq 1).$$

Taking $k = 6/d$ yields

$$\frac{6}{d}v = (3, 0), \quad v = (d/2, 0).$$

Thus d is even, so $d = 2$ or $d = 6$. Evaluation on the full final root box gives

$$-0.010319188238 < (F^2(p) - p)_2 < -0.010319188237. \quad (12)$$

If $d = 2$, the forced displacement is $v = (1, 0)$, contrary to (12). Hence $d = 6$.

For $n = 6k + r$, $0 \leq r < 6$, equivariance gives

$$F^n(p) - p = k(3, 0) + F^r(p) - p.$$

The final term takes only six values, so division by n and passage to the limit prove

$$\rho(F, p) := \lim_{n \rightarrow \infty} \frac{F^n(p) - p}{n} = (1/2, 0).$$

3.3. Ellipticity and low-order nonresonance. Let $M = DP(p) = DF^6(p)$ and $T = \text{tr } M$. The validated enclosure is

$$-1.991309 < T < -1.991308. \quad (13)$$

Since $\det M = 1$ exactly, the multipliers are

$$\lambda = \frac{T}{2} + is, \quad \bar{\lambda} = \frac{T}{2} - is, \quad s = \sqrt{1 - T^2/4} > 0. \quad (14)$$

They are nonreal and lie on the unit circle, which is the elliptic case of the trace-determinant classification [KH95]. A root of unity of order at most four would give trace in $\{2, -2, -1, 0\}$. Since (13) lies strictly between -2 and -1 , these possibilities are excluded. This proves the required finite-order nonresonance, not nonresonance at every order.

4. VALIDATION OF THE FIRST BIRKHOFF COEFFICIENT

4.1. Taylor jets on the entire root box. For a base point $z_0 \in X_{10^{-60}}$, write the ordinary third-order Taylor polynomial of $F^6(z_0 + \xi)$ in $\xi = (\xi_1, \xi_2)$. The coefficient of $\xi_1^j \xi_2^k$ is the corresponding derivative divided by $j!k!$. Every such coefficient is enclosed uniformly for all z_0 in the box. Translation by $(3, 0)$ does not change the positive-degree coefficients of P .

The polynomial algebra is truncated after total degree three. If $t = t_0 + \delta$ with δ having zero constant term, the exact jet identity is

$$\sin(t) = \sin(t_0) + \cos(t_0)\delta - \frac{1}{2}\sin(t_0)\delta^2 - \frac{1}{6}\cos(t_0)\delta^3 \pmod{\text{degree} \geq 4}. \quad (15)$$

Applying (15) to the two shears at each of the six steps produces interval enclosures of all required derivatives. At the exact root p , the constant term of $P(p + \xi) - p$ is zero. Ignoring the interval enclosure of this constant term when forming the local germ is therefore justified by Proposition 3.2, not by treating the center c as an exact fixed point.

4.2. An explicit complex eigenbasis. Write

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad q = \begin{pmatrix} B \\ \lambda - A \end{pmatrix}, \quad \xi = qz + \bar{q}\bar{z}.$$

The exact identity $\det M = 1$ implies $Mq = \lambda q$. The checker establishes $B > 0$ and $s > 0$, and

$$\det[q, \bar{q}] = -2iBs \neq 0.$$

The transformed Taylor polynomial of the local return map is therefore

$$z' = \lambda z + a_{20}z^2 + a_{11}z\bar{z} + a_{02}\bar{z}^2 + \sum_{j+k=3} a_{jk}z^j\bar{z}^k + O(|z|^4). \quad (16)$$

The coefficients in (16) are ordinary Taylor coefficients in this particular eigenbasis.

4.3. Quadratic elimination and the resonant coefficient. As explained in Section 1, for a map a monomial $z^j\bar{z}^k$ in the first complex component is resonant when

$$\lambda^j \bar{\lambda}^k = \lambda, \quad \text{equivalently} \quad \lambda^{j-k-1} = 1.$$

The low-order nonresonance proved above permits the removal of every quadratic term and of the nonresonant cubic terms. Set

$$z = u + h_{20}u^2 + h_{11}u\bar{u} + h_{02}\bar{u}^2 + O(|u|^3),$$

where comparison of quadratic terms gives

$$h_{20} = \frac{a_{20}}{\lambda^2 - \lambda}, \quad h_{11} = \frac{a_{11}}{1 - \lambda}, \quad h_{02} = \frac{a_{02}}{\bar{\lambda}^2 - \lambda}. \quad (17)$$

All denominators are separated from zero by the interval calculation.

Let $\mathcal{P}$ denote the germ in (16), and let H denote the coordinate change. In the conjugacy equation $\mathcal{P} \circ H = H \circ N$, the cubic contribution from the quadratic terms is $D\mathcal{P}_2 H_2$. Its coefficient of $u^2 \bar{u}$ gives

$$b_{21} = a_{21} + 2a_{20}h_{11} + a_{11}h_{20} + a_{11}\bar{h}_{11} + 2a_{02}\bar{h}_{02}. \quad (18)$$

A cubic coordinate change contributes a multiple of $\lambda - \lambda^2 \bar{\lambda} = 0$ to this resonant monomial. Thus it does not change b_{21} . In particular, b_{21} is *not* the raw Taylor coefficient a_{21} .

As an algebraic cross-check, the verifier also evaluates

$$\frac{b_{21}}{\lambda} = \frac{a_{21}}{\lambda} + \frac{(1 - 2\lambda)a_{20}a_{11}}{\lambda^2(\lambda - 1)} + \frac{|a_{11}|^2}{\lambda - 1} + \frac{2|a_{02}|^2}{\lambda^3 - 1}. \quad (19)$$

This follows from (17)–(18) and $\bar{\lambda} = \lambda^{-1}$. The two independently evaluated interval expressions overlap, as required by the exact identity.

Area preservation implies

$$\operatorname{Re}(b_{21}/\lambda) = 0. \quad (20)$$

One way to see this without assuming the polynomial change H is exactly symplectic is to use the area density preserved by the conjugated map. After quadratic and nonresonant cubic elimination, this map has the form

$$N(u) = \lambda u + b_{21}u^2 \bar{u} + O(|u|^4).$$

The preserved density has a nonzero constant term. Its linear term vanishes because invariance would make that term invariant under a nontrivial rotation. At quadratic order, the coefficient of $|u|^2$ in the density difference under the rotation is zero, whereas the same coefficient in the Jacobian determinant of N is $4 \operatorname{Re}(b_{21}/\lambda)$. Density invariance gives (20). Equivalently, the unique quadratic homological solution and the invariant resonant cubic coefficient agree with those obtained by symplectic Birkhoff normalization.

Define $\beta = \operatorname{Im}(b_{21}/\lambda)$. The checker proves

$$600142873429.5456140196 < \beta < 600142873429.5456140198. \quad (21)$$

The computed enclosure of $\operatorname{Re}(b_{21}/\lambda)$ is contained in $(-4 \cdot 10^{-35}, 4 \cdot 10^{-35})$ and contains zero. This is a consistency check; the exact equality in (20) follows analytically from area preservation.

4.4. Symplectic normalization and its sign. The real matrix taking $(\operatorname{Re} z, \operatorname{Im} z)$ to ξ is

$$S = [2 \operatorname{Re} q, -2 \operatorname{Im} q], \quad \det S = -4Bs.$$

The validated determinant satisfies

$$-3.567294748 < \det S < -3.567294747. \quad (22)$$

Set $d = -\det S > 0$ and define

$$\zeta = \sqrt{d}\bar{z} = Q + iR. \quad (23)$$

The map from (Q, R) to ξ has matrix

$$S \begin{pmatrix} 1/\sqrt{d} & 0 \\ 0 & -1/\sqrt{d} \end{pmatrix} \quad \text{and determinant} \quad -\frac{\det S}{d} = 1,$$

so it is symplectic for $dQ \wedge dR$ and $d\xi_1 \wedge d\xi_2$. Complex conjugation in (23) is essential: scaling alone would leave the orientation negative. The linear multiplier in the new coordinate is $\mu = \bar{\lambda}$.

In the convention where the symplectically normalized complex coordinate is $w = Q + iR + O(|(Q, R)|^2)$ and the Birkhoff normal form reads

$$w' = \mu w + i\tau_{\text{can}}\mu w|w|^2 + O(|w|^4), \quad (24)$$

we obtain

$$\tau_{\text{can}} = -\frac{\beta}{d} = \frac{\beta}{\det S}. \quad (25)$$

Indeed, conjugating the resonant normal form and scaling by $\sqrt{d}$ changes the coefficient $i\beta\lambda$ to $-i(\beta/d)\bar{\lambda}$. Equations (21)–(25), evaluated with full interval endpoints, give (7).

The verifier independently repeats the Taylor-coordinate calculation in the normalized eigenbasis $\bar{q}/\sqrt{d}$, with multiplier $\bar{\lambda}$, and obtains the same displayed enclosure for the first Birkhoff coefficient. Hence the elliptic fixed point is twist-nondegenerate.

5. THE LOCAL KAM CONCLUSION

We use the following standard local consequence of Moser's invariant-curve theorem [Mos62, Mos]; the positive-measure formulation near a generic elliptic point is also recalled in [LM15].

Theorem 5.1 (Local invariant curves near a twist-nondegenerate elliptic point). *Let a real-analytic area-preserving surface diffeomorphism have an elliptic fixed point. Suppose its multipliers satisfy $\lambda^j \neq 1$ for $1 \leq j \leq 4$ and its first Birkhoff coefficient is nonzero. Then arbitrarily small neighborhoods of the fixed point contain a positive-area measurable union of smooth invariant Jordan curves surrounding the point. The restriction to each selected curve is smoothly conjugate to a Diophantine rigid rotation.*

Here a circle rotation number $\omega \in \mathbb{R}/\mathbb{Z}$ is Diophantine in the sense of (4). This condition concerns the rotations on the invariant curves, not the multiplier angle at the central fixed point.

The hypotheses of Theorem 5.1 hold for P at p : analyticity and area preservation follow from the shear decomposition, ellipticity and low-order nonresonance from (13), and nonzero twist from (7). On a sufficiently small coordinate disk around $\bar{p}$, the projected local dynamics are those of $g = f^6$. We therefore obtain the set K in Theorem 2.1.

The local intersection property required by the annular form of the invariant-curve theorem is automatic here. A sufficiently small Jordan curve surrounding p and its image under

P enclose the same area and both enclose p . If they were disjoint, one enclosed domain would be properly contained in the other, contradicting equality of area. Equivalently, on a simply connected plane neighborhood, the closed one-form $P^*(x dy) - x dy$ is exact.

A single invariant Jordan curve also bounds a P -invariant open disk: a plane homeomorphism preserving that curve preserves its bounded complementary component. Such disks, and their six projected images, justify the term *elliptic island*. This does not mean that the entire disk is foliated by KAM curves. The pointwise zero-exponent assertion below concerns the selected KAM set and its saturation.

6. POINTWISE ZERO EXPONENTS AND NON-ERGODICITY

6.1. A derivative-cocycle lemma.

Lemma 6.1. *Let g be a C^1 area-preserving surface diffeomorphism. Work in an area coordinate chart, so that the preserved area form is $dx \wedge dy$. Suppose a g -invariant curve admits a C^1 embedding $h : \mathbb{T}^1 \rightarrow \mathbb{R}^2$ in this chart, with $h'(\theta) \neq 0$, such that*

$$g \circ h = h \circ R_\omega, \quad R_\omega(\theta) = \theta + \omega.$$

Then, at every point of the curve, both the derivative iterates and their inverses grow at most linearly. In particular, the forward Lyapunov exponent exists and is zero in every nonzero tangent-space direction.

Proof. Let $J(a, b) = (-b, a)$ be the rotation by $\pi/2$ and define

$$u(\theta) = h'(\theta), \quad v(\theta) = \frac{Ju(\theta)}{\|u(\theta)\|^2}, \quad B(\theta) = [u(\theta), v(\theta)].$$

Then $\det B(\theta) = 1$ because it determines a positively oriented rectangle whose dimensions are $\|u(\theta)\|$ and $\|u(\theta)\|^{-1}$. Differentiating the conjugacy gives

$$Dg(h(\theta))u(\theta) = u(\theta + \omega).$$

Thus the first column of $B(\theta + \omega)^{-1}Dg(h(\theta))B(\theta)$ is $(1, 0)^\top$. Its determinant is one, so

$$B(\theta + \omega)^{-1}Dg(h(\theta))B(\theta) = \begin{pmatrix} 1 & a(\theta) \\ 0 & 1 \end{pmatrix} \quad (26)$$

for a continuous function a . Multiplying these matrices along the orbit yields

$$B(\theta + n\omega)^{-1}Dg^n(h(\theta))B(\theta) = \begin{pmatrix} 1 & \sum_{k=0}^{n-1} a(\theta + k\omega) \\ 0 & 1 \end{pmatrix}. \quad (27)$$

The functions a , B , and B^{-1} are bounded on the circle. The inverse of the matrix on the right of (27) has the negative of the same upper-right entry. Hence, for some constant $C_h \geq 1$ and every $n \geq 0$,

$$\|Dg^n(h(\theta))\| + \|(Dg^n(h(\theta)))^{-1}\| \leq C_h(1 + n). \quad (28)$$

Consequently, for every $v_0 \neq 0$,

$$\frac{\|v_0\|}{C_h(1+n)} \leq \|Dg^n(h(\theta))v_0\| \leq C_h(1+n)\|v_0\|.$$

Taking logarithms, dividing by n , and letting $n \rightarrow \infty$ proves the assertion at every point. No appeal to an almost-everywhere regularity theorem is needed. $\square$

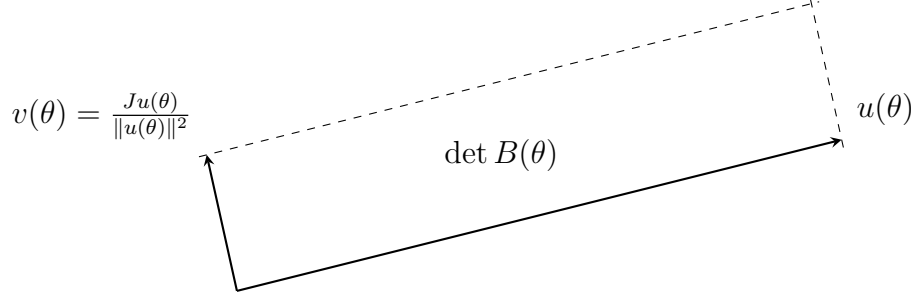


FIGURE 1. $u(\theta)$ and $v(\theta)$ enclose a rectangle of unit area, $\det B(\theta)$.

Apply Lemma 6.1 on every selected KAM curve. A uniform constant across different curves is unnecessary. For $n = 6q + r$, $0 \leq r < 6$,

$$Df^n(z) = Df^r(f^{6q}z) Dg^q(z).$$

The derivatives of the finitely many maps f^r , and their inverses, are uniformly bounded on the compact torus. The two-sided polynomial bounds therefore imply (8) for all $z \in K$. The identity

$$Df^n(f^j z) = Df^j(f^n z) Df^n(z) (Df^j(z))^{-1}$$

transfers the same conclusion to $f^j(K)$ for $0 \leq j \leq 5$. This proves the pointwise assertion on $\widehat{K}$.

6.2. An invariant set of intermediate measure. Choose the neighborhood U of $\bar{p}$ small enough that $m(U) < 1/6$, and choose the positive-area set $K \subset U$ supplied by Theorem 5.1. Since $f^6(K) = K$,

$$f(\widehat{K}) = \widehat{K}, \quad 0 < m(K) \leq m(\widehat{K}) \leq \sum_{j=0}^5 m(f^j K) = 6m(K) \leq 6m(U) < 1.$$

This invariant measurable set of intermediate measure proves that f is not ergodic. Since both exponents vanish on $\widehat{K}$, which has positive measure, the property of having two nonzero Lyapunov exponents almost everywhere also fails. This completes the proof of Theorem 2.1.

7. SCOPE AND QUALITATIVE PERSISTENCE

The result does not exclude positive-measure hyperbolic components elsewhere, and it does not claim that the complement of the KAM set is ergodic. It also provides no explicit numerical lower bound for the area of the invariant curves' union or of the enclosing island. The finite computation verifies the hypotheses of a local KAM theorem; it does not compute parametrizations of the invariant curves.

There is a straightforward qualitative extension to nearby parameters.

Corollary 7.1. *For all (K, L) in some open neighborhood of $(1, 1)$, the map $\Phi_{K,L}$ has a twist-nondegenerate elliptic six-cycle with lift rotation vector $(1/2, 0)$ and is not Lebesgue ergodic.*

Proof. Consider $G_{K,L}(z) = F_{K,L}^6(z) - z - (3, 0)$. At the validated root,

$$\det D_z G_{1,1}(p) = \det(M - I) = 2 - \operatorname{tr} M > 0.$$

The implicit function theorem gives a nearby periodic point depending continuously, indeed analytically, on (K, L) . The strict trace and twist inequalities persist by continuity. The period-two displacement remains nonzero, so the same integer-displacement argument gives least period six. Moser's theorem and the invariant-saturation argument apply to every such nearby parameter. $\square$

No explicit size for this parameter neighborhood is claimed. Obtaining one would require a parameter-dependent interval calculation.

APPENDIX A. ARITHMETIC, CERTIFICATES, AND REPRODUCIBILITY

A.1. Dyadic intervals. The ancillary file `validate_island.py` uses Python's arbitrary-precision integers and the standard library only. At precision b , a real interval is stored as

$$[\ell/2^b, u/2^b], \quad \ell, u \in \mathbb{Z}.$$

A rational a/d is enclosed by integer floor and ceiling of $2^b a/d$. Addition and subtraction of endpoint numerators are exact. Multiplication takes the minimum and maximum of the four endpoint products and rounds the quotients by 2^b outward. Division, allowed only when the denominator interval excludes zero, similarly bounds the four endpoint ratios. Square-root endpoints are obtained from integer square roots of $\ell 2^b$ and $u 2^b$, increasing the upper integer when necessary. Complex intervals are rectangles with these real intervals as components.

A.2. Transcendental enclosures. An interval for π follows from Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

For each arctangent, the alternating series has decreasing terms and its tail is bounded by the first omitted term. All partial sums and tail bounds are enclosed outward.

For sine and cosine, subtract an integer multiple of the interval for 2π to obtain a small argument interval. The integer may be selected from interval midpoints because the

trigonometric identity holds for every integer choice. At a dyadic midpoint t of the reduced interval, use

$$S_{j-1}(t) = \sum_{k=0}^{j-1} \frac{(-1)^k t^{2k+1}}{(2k+1)!}, \quad C_{j-1}(t) = \sum_{k=0}^{j-1} \frac{(-1)^k t^{2k}}{(2k)!}.$$

Taylor's theorem, including the intervening zero coefficient, gives

$$|\sin t - S_{j-1}(t)| \leq \frac{|t|^{2j+1}}{(2j+1)!}, \quad |\cos t - C_{j-1}(t)| \leq \frac{|t|^{2j}}{(2j)!}.$$

The checker encloses these bounds with interval recurrences. If δ is the maximum distance from the midpoint to an endpoint, adding $[-\delta, \delta]$ extends either enclosure to the entire argument interval, since both sine and cosine are 1-Lipschitz on $\mathbb{R}$. Thus no library floating-point trigonometric value enters a proof test.

A Sage notebook available. A Sage notebook [Las26] is included as a source for the orbit data and the original root, period, and trace validation. The self-contained checker distributed with this manuscript additionally supplies the complete third-order jet and Birkhoff-coefficient calculation. Reproducing the present certificate requires only the accompanying files, although direct check at [Las26] would be helpful.

A.3. Execution and outputs. From the directory containing the checker, run

```
python3 validate_island.py --bits 256 --output certificate_256.json
python3 validate_island.py --bits 384 --output certificate_384.json
python3 validate_island.py --bits 512 --output certificate_512.json
python3 test_validator.py
```

The recorded runs use Python 3.13.5. The checker uses syntax compatible with Python 3.10 and later. Each successful validation run ends with `ALL CHECKS PASSED`. Failure of any proof condition raises an exception; these checks remain active under `python -O`.

The JSON files record the integer endpoint numerators, their common denominator, the exact input center and preconditioner, the root certificates, the real Taylor jets, the complex normal-form coefficients, and a SHA-256 digest of the checker source. The accompanying decimal displays are rounded outward but are not used to decide inequalities. The logs record the full execution output at all three precisions.

Besides all strict inequalities displayed in the paper, the checker compares the linear Taylor jet with a separate chain-rule derivative computation, evaluates both formulas (18) and (19), and recomputes the twist in the symplectically normalized eigenbasis. The diagnostic test file checks rational interval operations, square roots, trigonometric identities, ordinary Taylor coefficient conventions, and recovery of a known twist after a nonlinear symplectic conjugacy.

These consistency checks and repeated precisions are useful diagnostics, not substitutes for interval inclusion. The proof rests on the inclusion-preserving arithmetic, the finite strict inequalities, and the analytic arguments above. Our work is a computer-assisted proof.

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SCHOOL OF MATHEMATICAL AND STATISTICAL SCIENCES, CLEMSON UNIVERSITY, SC 29631
 Email address: elasa@clemson.edu